



OCT 2024

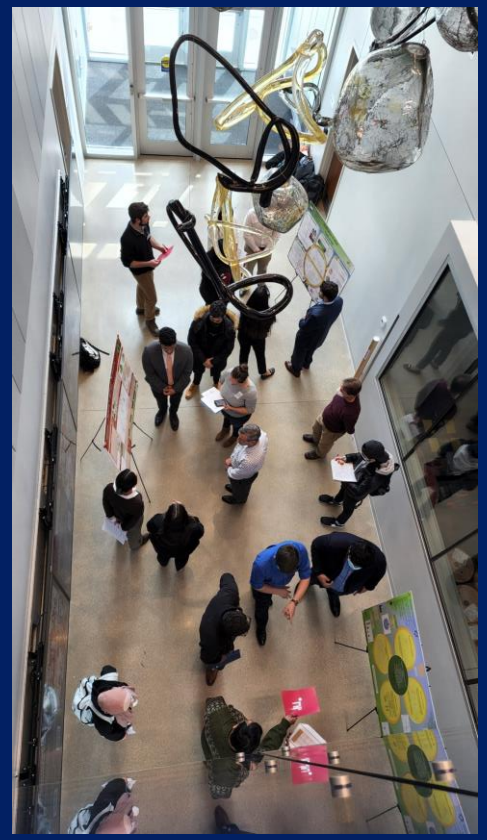
# Department of Chemical Engineering

Vikas Berry

Department Head and Satish C. & Asha Saxena Professor

UIC UNIVERSITY OF ILLINOIS CHICAGO

DEPARTMENT UPDATES



# 43rd Academic Year as Chemical Engineering at UIC

# STUDENTS AND FACULTY

Faculty Members : 16 (13.25 FTE)
National Award Winners // CEOs // Industry Experts

Tenure-Track : 11
- Full : 4
- Associate : 5
- Assistant : 2
- % Female : 18.18%
- Bridge-2-Faculty: 1

Clinical-Track
- Assistant : 1
- Associate : 3
- % Female : 25%

Research-Track : 1

Affiliated Faculty : 18

BS Degree Program : 165 students
Graduate Program : 55 students
- MS Degree Program : 13 students
- PhD Degree Program : 42 students

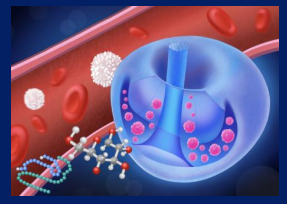
BS-MS Joint Program : 10 students

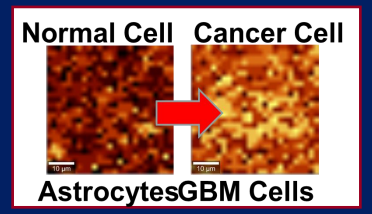


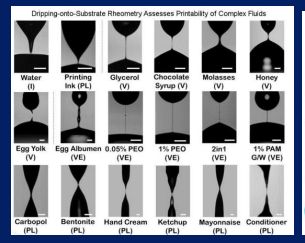
## Diversity Statement

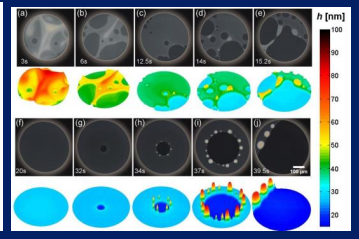
UIC chemical engineering fosters an inclusive environment that encourages, supports, and empowers diverse voices, ideas, and opinions. We value equity, inclusion, and all forms of diversity within our faculty, staff, and student population. Our multicultural, pluralistic community brings richness in experience, backgrounds, perspectives, and skills to all activities in the department.

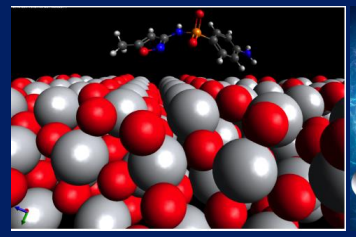
# STAFF

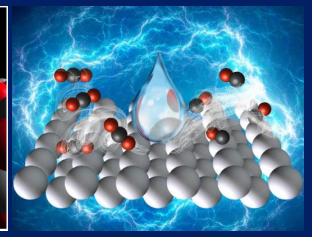


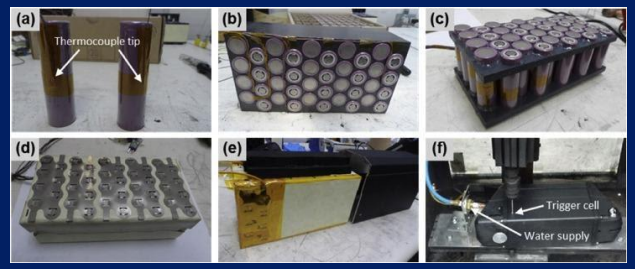







**Sarai Chavez** – Office Support Assistant

**Meghan King** – Assistant Director of Academic Communications

**Eduardo Martinez** – Grad./Undergraduate Program Coordinator

**Jan Raphael Sagun** – Laboratory and Computers Manager

**Cassaundra Wilson** – Business Manager

**COE (Monica Tith)** – HR

DEPARTMENT UPDATES

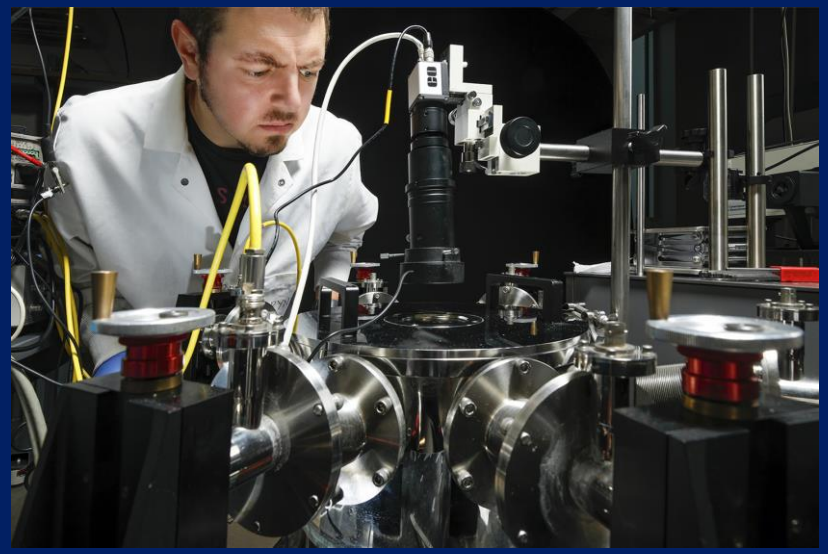
# Rankings

- US-NEWS
  - Chemical Engineering (overall) = 62
  - Chemical Engineering (public) = 38

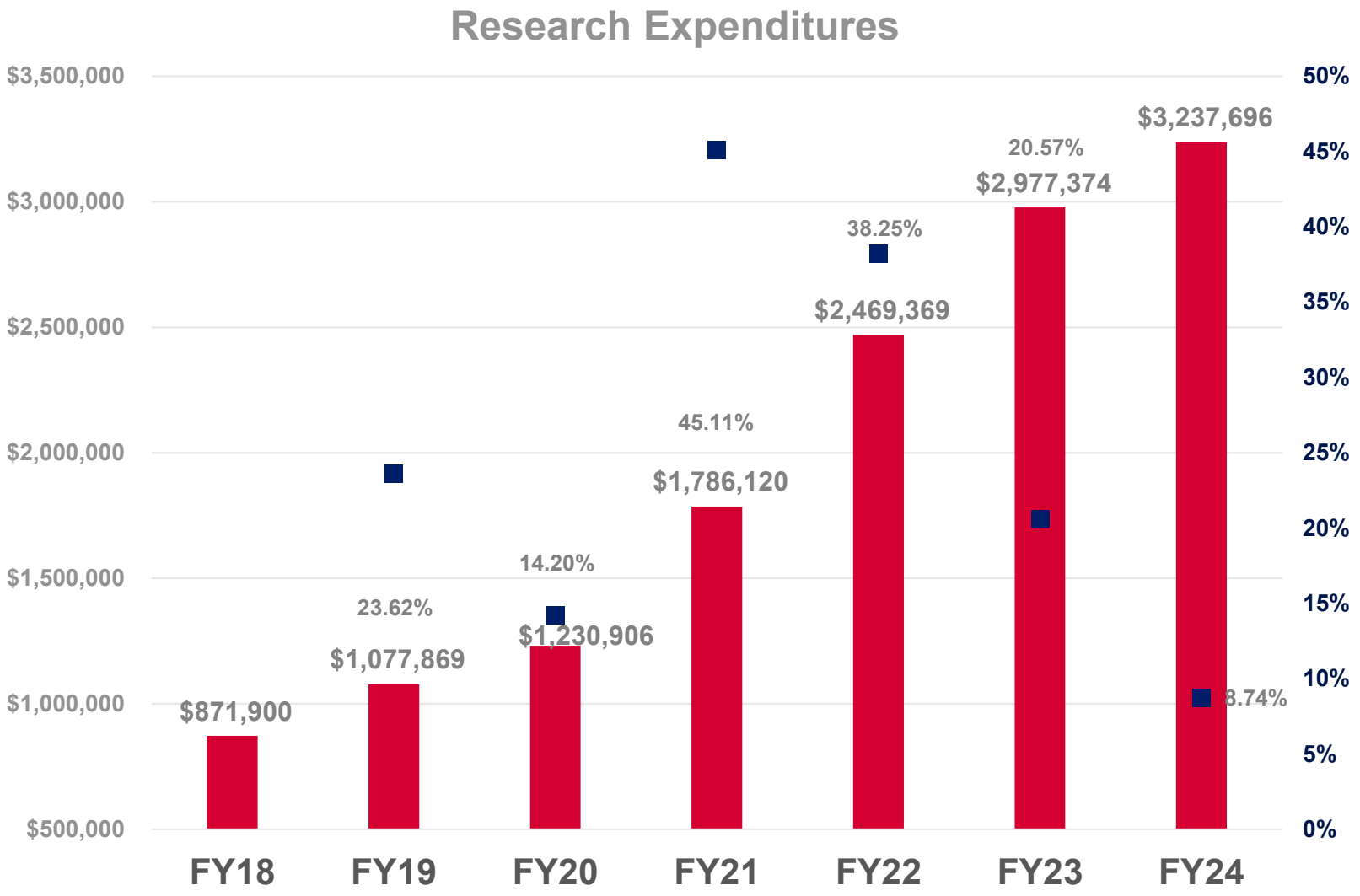
- QS Ranking
  - Chemical Engineering (overall) = 35
  - Chemical Engineering (public) = 22

DEPARTMENT UPDATES

# Expenditures



## Research Expenditures:

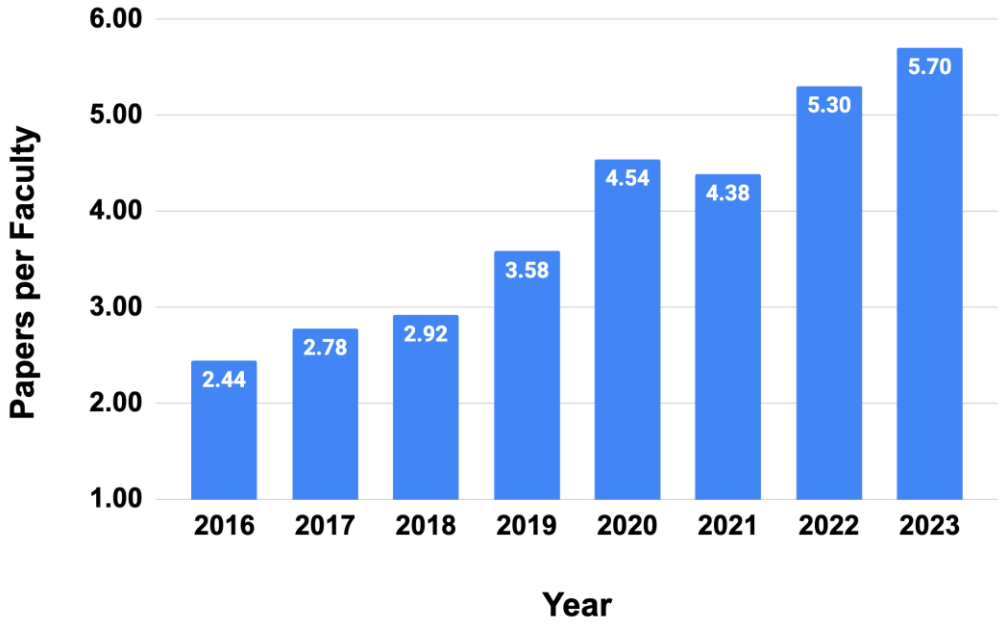


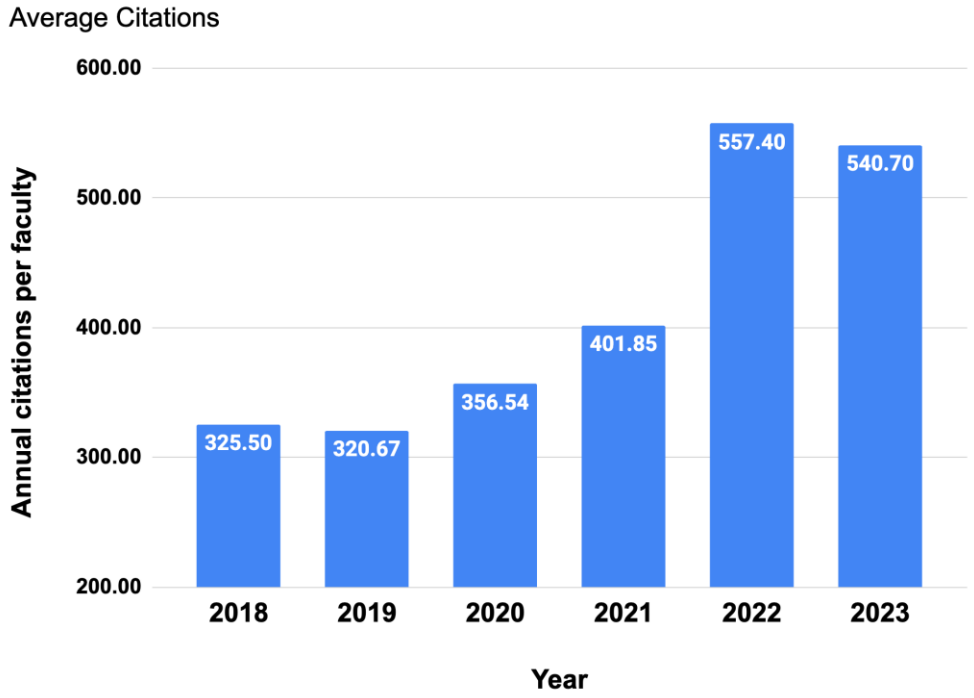
RESEARCH PRODUCTIVITY

# Publications

**Tot Citations/Fac = 3971**

**Avg. h-Index = 24.31**

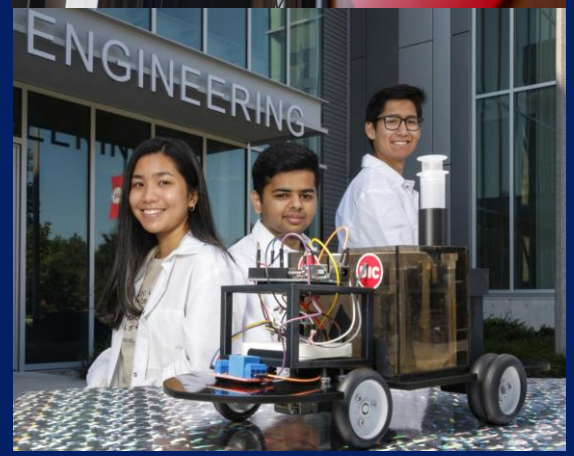


# KEY ACCOMPLISHMENTS



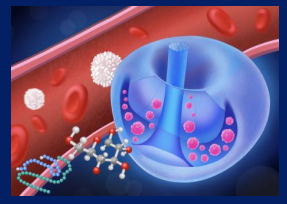


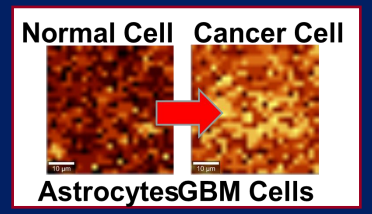



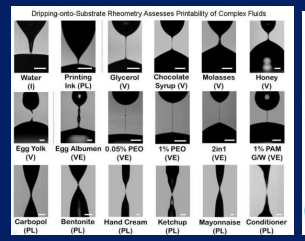

# STUDENT PLACEMENTS

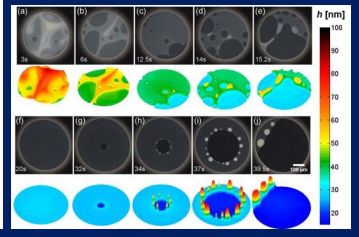
Honeywell UOP
DUPONT
TESLA
DOW
L'ORÉAL
bp
Exxon
Ingredion
BAKERRISK
tpi TheraProteins
JACOBS
NALCO
CABOT
GELITA Improving Quality of Life
LA-CO
Abbott
AthenaVisual, Inc.
Ambitech
GATX
CBI
NAVISTAR
OROCHEM Molecular Purification by Design
ENengineering
EXIDE TECHNOLOGIES
SCP South Chicago Packing
CARDINAL INTELLECTUAL PROPERTY
Air Liquide
Cornerstone Controls 75 Years
middough
Sargent & Lundy LLC

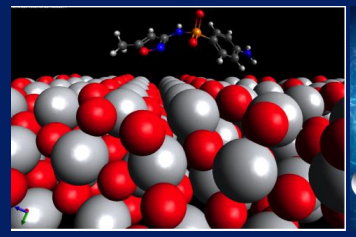
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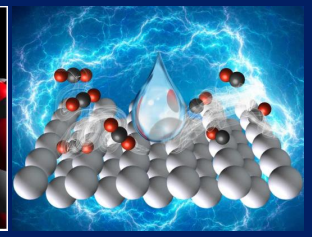


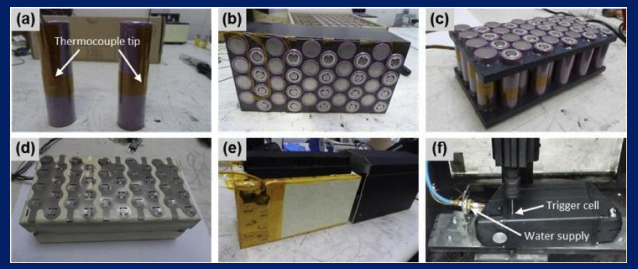







## Chemical Engineering is Sustainability

- Energy
  - Battery Research
  - Sustainable Energy Technologies
- Water
  - Water Resource Sustainability
- Human Health
  - Therapeutics (Drug Delivery)
  - Diagnostics
- Food
- Materials

- Funding in force: $15M

WORLD-CLASS FACULTY MAKES A WORLD-CLASS DEPARTMENT



# The Stanford Top 2% researchers rankings report

**Career rankings from 1960:**

- Professor Vikas Berry
- Professor Brian Chaplin
- Research Professor Said Al-Hallaj

**Citations and publications for 2022:**

- Professor Brian Chaplin
- Professor Vikas Berry
- Research Professor Said Al-Hallaj
- Associate Professor Meenesh Singh
- Associate Professor Vivek Sharma

WORLD-CLASS FACULTY MAKES A WORLD-CLASS DEPARTMENT

# Entrepreneurship

Chemical Engineering Faculty and Students are collaborating with the following resources to translate ideas into products

MHUB

DPI DISCOVERY PARTNERS INSTITUTE

MATTER

UIC Innovation Center

NSF I-CORPS HUB GREAT LAKES REGION UIC INNOVATION

CHANCELLOR'S TRANSLATIONAL RESEARCH INITIATIVE UIC INNOVATION

13 faculty/student startups
3 CTRI grants received
2 POC grants received
1 DPI grant received
3 NSF-ICORPs
2 operations in mHUB

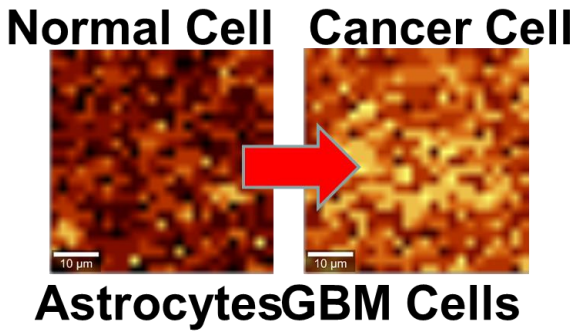


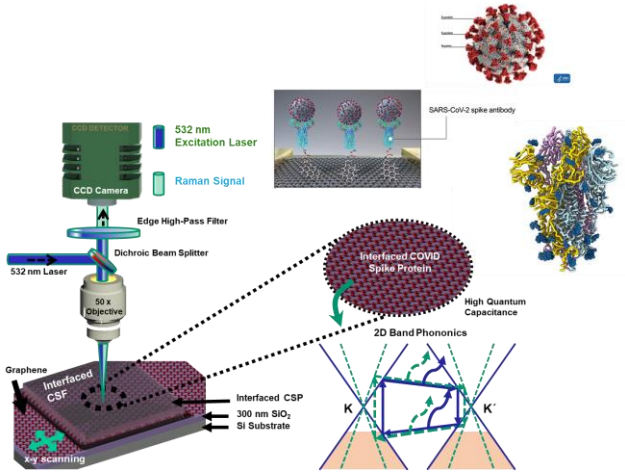


UIC

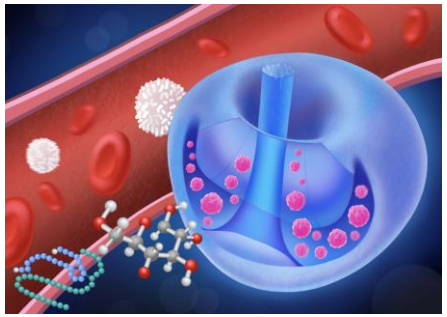
# SOME RESEARCH AREAS

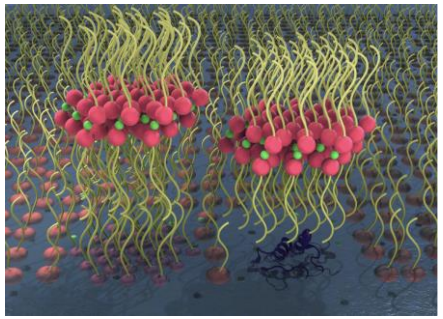
- **COVID and Cancer Detection**

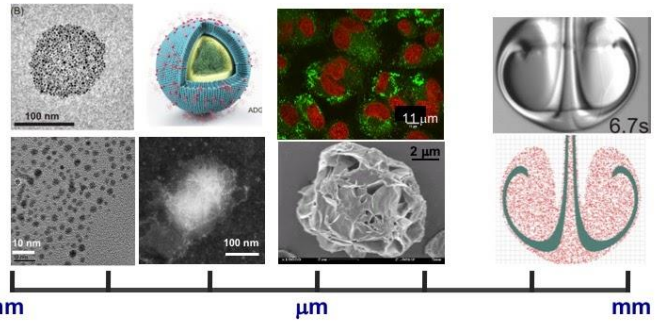


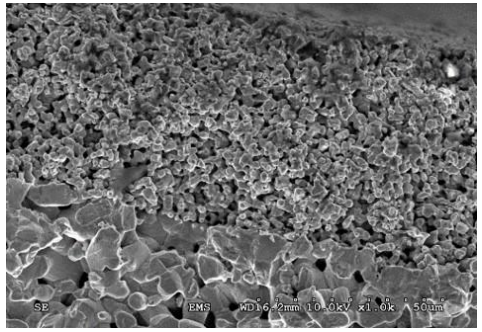

- **Drug Delivery for Vaccine Development**








- **Water**





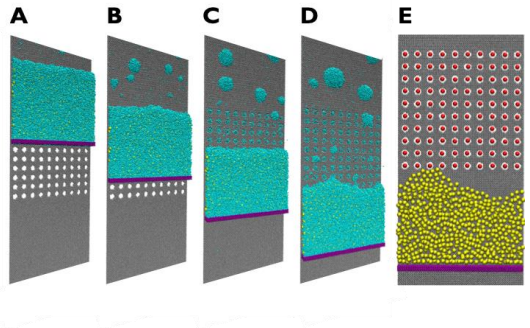

Engineering/Chemical Engineering/

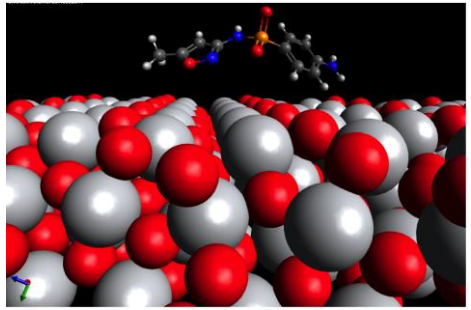
# SOME RESEARCH AREAS

- **Bioenergy and Sustainability**

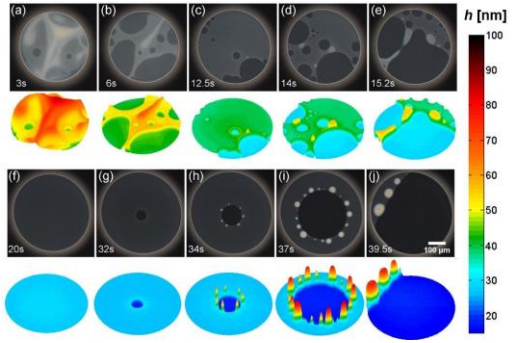



- **Simulation of Self-Assembly in Nanotechnology**

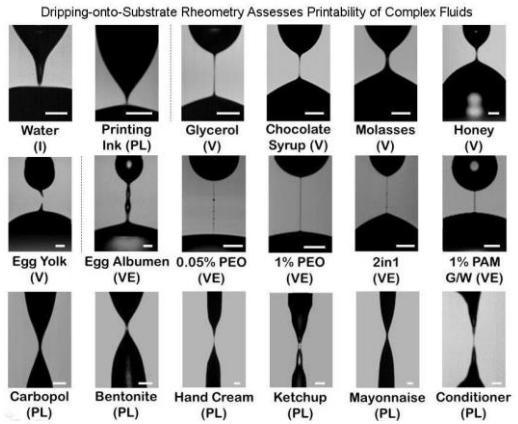


- **Printability of Complex Fluids**

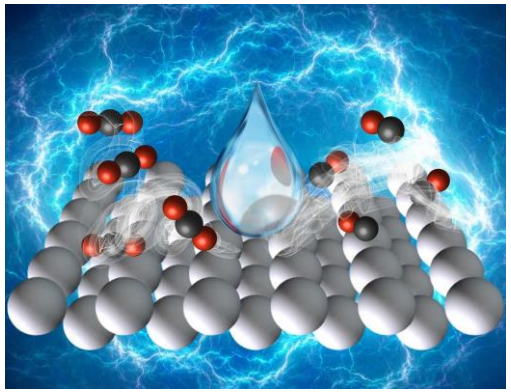



Engineering/Chemical Engineering/

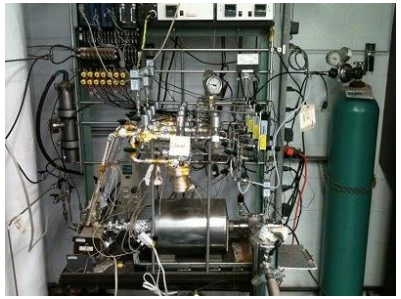


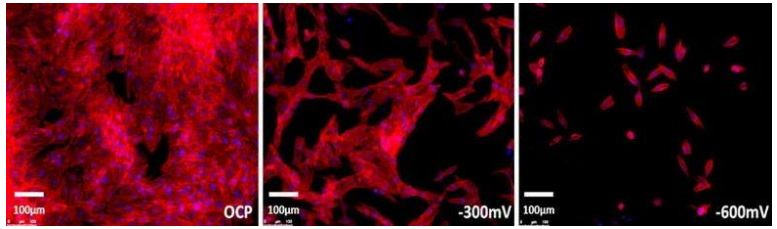
# SOME RESEARCH AREAS

- **Artificial Photosynthesis, Crystallization**

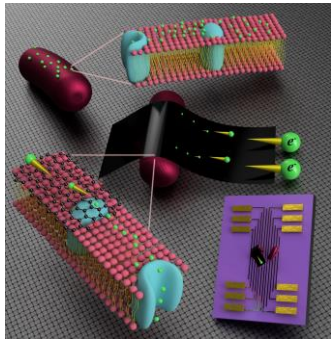


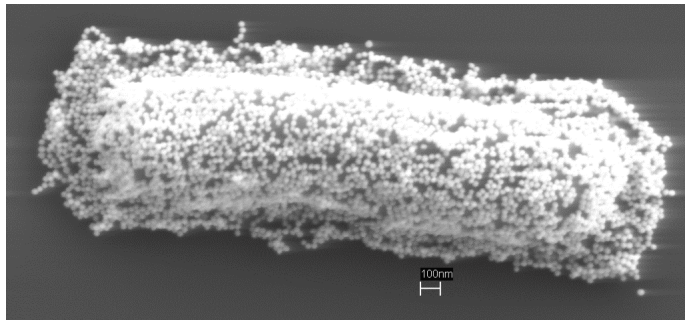
- **Microelectronics Processing**






- **Microbial Fuel Cells**

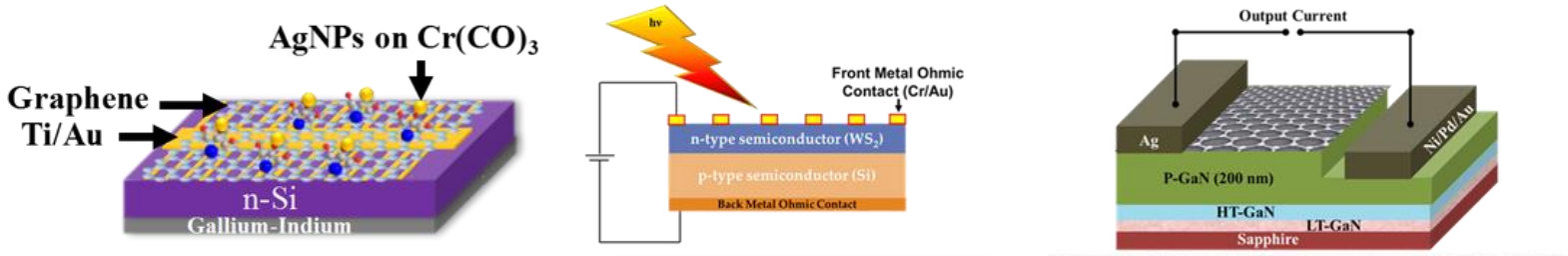




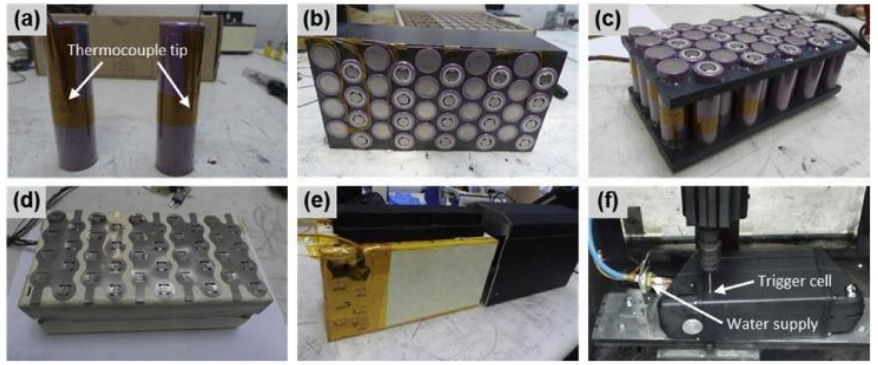

Engineering/Chemical Engineering/

# SOME RESEARCH AREAS

- **Few-Atoms-Thick Solar Cells**




- **Thermal Batteries**





OCTOBER 1, 2024

# Town Hall Meeting: ChE Undergraduate Program

Alan Zdunek
Clinical Associate Professor
Director of Undergraduate Studies

UIC UNIVERSITY OF ILLINOIS CHICAGO

# ABET Program Educational Objectives

| PEO | Category | Description |
|---|---|---|
| 1 | Depth | In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation. |
| 2 | Breadth | Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education. |
| 3 | Professionalism | In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment. |
| 4 | Life-long learning | Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership. |

# ChE Curriculum:

## Core Courses

**Thermodynamics-Based Courses**
- ChE 201 – Introduction to thermodynamics
- ChE 301 – Chemical Engineering Thermodynamics
- ChE 321 – Chemical Reaction Engineering

**Transport-Based Courses**
- ChE 311 – Transport Phenomena I (Fluid Mechanics)
- ChE 312 – Transport Phenomena Ii (Heat & Mass Transfer)
- ChE 313 – Transport Phenomena III (Separations)

**Process-Oriented Courses**
- ChE 210 – Material & Energy Balances
- ChE 341 – Chemical Process Control
- ChE 396 – Senior Design I (Equipment Design)
- ChE 397 – Senior Design II (Design teams)

**Unit Operations Laboratory**
- ChE 381 – Chem Eng Lab I
- ChE 382 – Chem Eng Lab II

**Molecular Engineering Courses**
- CHE 230 Molecular Engineering in CHE
- CHE 330 Polymer Science

**Numerical Analysis**
- ChE 205- Numerical Methods in ChE

**Engineering Education & Development**
- ENGR100 -Engineering Orientation (Engineering Success Program, Guaranteed Paid Internship Program)
- ChE 499- Professional Development Seminar

ChE 400-level Elective
CHE 392: Undergraduate Research

## ***Six Concentrations (15% of ChE students):***

| | |
|---|---|
| **Biochemical Engineering** | **9 students** |
| **Process Automation** | **6** |
| **Energy & Environment** | **4** |
| **Nanotechnology** | **3** |
| **Polymers & Molecular Engineering** | **3** |
| **Entrepreneurship** | **0** |

# Curriculum Improvement in progress…

Chemical engineers are at the forefront of transforming our future with sustainable solutions

## PROGRAM REVISION IDEAS/ PROPOSALS:

(1)

ADD CHE 101 INTRODUCTION TO SUSTAINABLE CHEMICAL ENGINEERING TO CORE CURRICULUM (+3 HOURS)

CREATE A NEW SELECTIVE OPTION: CHE 230 MOLECULAR SYSTEMS IN CHEMICAL ENGINEERING OR CME 260 PROPERTIES OF MATERIALS (-3 HOURS)

(2)

REMOVE CHEMISTRY COURSE FROM MATH/SCIENCE REQUIREMENT (-3 HOURS)

ADD SECOND TECHNICAL ELECTIVE TO CORE REQUIREMENT (+3 HOURS)

(3)

ADD DATA SCIENCE TO CHE CURRICULUM:

CREATE A NEW DATA SCIENCE CONCENTRATION

# JOB OPPORTUNITIES









# Chemical Engineering Employment Sectors

biotecnika
Your Bio Resource

- Biotechnology
- Chemical and allied products
- Process Plant Construction
- Pharmaceutical & Toiletries
- Water Industry
- Food & Beverage industries
- Environment Protection & Recycling
- Energy (Oil, Gas, & Nuclear)
- Process plant & Equipment Manufacture
- Materials (Plastics, dyes, metals, paper, etc.)
- Academics
- IT
- Consultancy
- Others

# UIC Chemical Engineering graduates are in high demand




**Ana Hernandez** (She/Her) · 1st
Process Manager @L'Oréal | People Leader |



**Jordan Flanigan** · 1st
Semiconductor Process Engineer at Intel



**Aesha Talia** · 1st
Product Manager at Medline Industries, Inc.



**Emir Asani** · 1st
Continuous Improvement Engineer at Mars Ice Cream



**Courteney Bruce-Kotey** · 1st
Process Engineer at Brindley Engineering Corporation



**Paul Diaz** · 1st
System Manager for digital I&C and mechanical equipment at Constellation Nuclear

# Student - ChE Community Interaction

- WELCOME BACK PICNIC
- END-OF-YEAR BANQUET
- AICHE STUDENT CHAPTER

GENERAL MEETINGS, PROFESSIONAL & FUN ACTIVITIES

AICHE NATIONAL MEETING, NORTH-CENTRAL REGION

CHEME CAR, CHEME CUBE

- CHE EXPO
- UIC CHEMICAL ENGINEERING LINKEDIN GROUP

- PAST EVENTS:

CHE INDUSTRY DAY- PROFESSIONAL ENGINEER PANEL DISCUSSION

CHE FORUM:

'MINI-CONFERENCE' WITH TECHNICAL SESSIONS

ALUMNI 'MIXER': STUDENTS & FACULTY INVITED



OCTOBER 1, 2024

# Town Hall Meeting: ChE Graduate Program

Brian P. Chaplin
Professor
Director of Graduate Studies

UIC UNIVERSITY OF ILLINOIS CHICAGO

# ChE Graduate Curriculum:

## *MS in Chemical Engineering:*

- 4+1 Format: Two graduate classes senior year
- One-Year Format: ‘Fast Track’ in two semesters
- Two-Year Format:
  - Coursework-Only Option (9 courses)
  - Project Option (8 courses + project)
  - Thesis Option (6 courses + thesis)

## *PhD in Chemical Engineering:*

- 6 required core courses
- 6 electives
- Departmental Seminar
- 52-60 semester hours of dissertation research
- Publication requirement set by advisor

# Graduate Program Revisions under Consideration…

***Proposed Revisions:***

- Adding a professional development course
- Reduce number of required courses and increase elective options
- Develop a PhD checklist to self assess progress
- Institute regular formal student assessments by advisor
- Standardize the prelim timing/content
- Provide timely decisions on TA/RA appointments
- Update student handbook

# Discussion Points…

***Survey Results:***

- **Need for updated graduate student handbook and structured PhD program guidelines**
- **Issues with timely TA/RA letter submissions and funding stability**
- **Disconnect between research labs**
- **Concerns about advisor-student relationships, including unreasonable demands and micromanagement**
- **Conflict Resolution:**
  - **Advisor → DGS → Dept. Head → Graduate College**



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| 1 | Depth | In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation. |
| 2 | Breadth | Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education. |
| 3 | Professionalism | In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment. |
| 4 | Life-long learning | Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership. |

# ChE Curriculum:

## Core Courses

**Thermodynamics-Based Courses**
- ChE 201 – Introduction to thermodynamics
- ChE 301 – Chemical Engineering Thermodynamics
- ChE 321 – Chemical Reaction Engineering

**Transport-Based Courses**
- ChE 311 – Transport Phenomena I (Fluid Mechanics)
- ChE 312 – Transport Phenomena Ii (Heat & Mass Transfer)
- ChE 313 – Transport Phenomena III (Separations)

**Process-Oriented Courses**
- ChE 210 – Material & Energy Balances
- ChE 341 – Chemical Process Control
- ChE 396 – Senior Design I (Equipment Design)
- ChE 397 – Senior Design II (Design teams)

**Unit Operations Laboratory**
- ChE 381 – Chem Eng Lab I
- ChE 382 – Chem Eng Lab II

**Molecular Engineering Courses**
- CHE 230 Molecular Engineering in CHE
- CHE 330 Polymer Science

**Numerical Analysis**
- ChE 205- Numerical Methods in ChE

**Engineering Education & Development**
- ENGR100 -Engineering Orientation (Engineering Success Program, Guaranteed Paid Internship Program)
- ChE 499- Professional Development Seminar

ChE 400-level Elective
CHE 392: Undergraduate Research

## ***Six Concentrations (15% of ChE students):***

| | |
|---|---|
| **Biochemical Engineering** | **9 students** |
| **Process Automation** | **6** |
| **Energy & Environment** | **4** |
| **Nanotechnology** | **3** |
| **Polymers & Molecular Engineering** | **3** |
| **Entrepreneurship** | **0** |

# Curriculum Improvement in progress…

Chemical engineers are at the forefront of transforming our future with sustainable solutions

## PROGRAM REVISION IDEAS/ PROPOSALS:

(1)

ADD CHE 101 INTRODUCTION TO SUSTAINABLE CHEMICAL ENGINEERING TO CORE CURRICULUM (+3 HOURS)

CREATE A NEW SELECTIVE OPTION: CHE 230 MOLECULAR SYSTEMS IN CHEMICAL ENGINEERING OR CME 260 PROPERTIES OF MATERIALS (-3 HOURS)

(2)

REMOVE CHEMISTRY COURSE FROM MATH/SCIENCE REQUIREMENT (-3 HOURS)

ADD SECOND TECHNICAL ELECTIVE TO CORE REQUIREMENT (+3 HOURS)

(3)

ADD DATA SCIENCE TO CHE CURRICULUM:

CREATE A NEW DATA SCIENCE CONCENTRATION

# JOB OPPORTUNITIES









# Chemical Engineering Employment Sectors

biotecnika
Your Bio Resource

- Biotechnology
- Chemical and allied products
- Process Plant Construction
- Pharmaceutical & Toiletries
- Water Industry
- Food & Beverage industries
- Environment Protection & Recycling
- Energy (Oil, Gas, & Nuclear)
- Process plant & Equipment Manufacture
- Materials (Plastics, dyes, metals, paper, etc.)
- Academics
- IT
- Consultancy
- Others

# UIC Chemical Engineering graduates are in high demand




**Ana Hernandez** (She/Her) · 1st
Process Manager @L'Oréal | People Leader |



**Jordan Flanigan** · 1st
Semiconductor Process Engineer at Intel



**Aesha Talia** · 1st
Product Manager at Medline Industries, Inc.



**Emir Asani** · 1st
Continuous Improvement Engineer at Mars Ice Cream



**Courteney Bruce-Kotey** · 1st
Process Engineer at Brindley Engineering Corporation



**Paul Diaz** · 1st
System Manager for digital I&C and mechanical equipment at Constellation Nuclear

# Student - ChE Community Interaction

- WELCOME BACK PICNIC
- END-OF-YEAR BANQUET
- AICHE STUDENT CHAPTER

GENERAL MEETINGS, PROFESSIONAL & FUN ACTIVITIES

AICHE NATIONAL MEETING, NORTH-CENTRAL REGION

CHEME CAR, CHEME CUBE

- CHE EXPO
- UIC CHEMICAL ENGINEERING LINKEDIN GROUP

- PAST EVENTS:

CHE INDUSTRY DAY- PROFESSIONAL ENGINEER PANEL DISCUSSION

CHE FORUM:

'MINI-CONFERENCE' WITH TECHNICAL SESSIONS

ALUMNI 'MIXER': STUDENTS & FACULTY INVITED



OCTOBER 1, 2024

# Town Hall Meeting: ChE Graduate Program

Brian P. Chaplin
Professor
Director of Graduate Studies

UIC UNIVERSITY OF ILLINOIS CHICAGO

# ChE Graduate Curriculum:

## *MS in Chemical Engineering:*

- 4+1 Format: Two graduate classes senior year
- One-Year Format: ‘Fast Track’ in two semesters
- Two-Year Format:
  - Coursework-Only Option (9 courses)
  - Project Option (8 courses + project)
  - Thesis Option (6 courses + thesis)

## *PhD in Chemical Engineering:*

- 6 required core courses
- 6 electives
- Departmental Seminar
- 52-60 semester hours of dissertation research
- Publication requirement set by advisor

# Graduate Program Revisions under Consideration…

***Proposed Revisions:***

- Adding a professional development course
- Reduce number of required courses and increase elective options
- Develop a PhD checklist to self assess progress
- Institute regular formal student assessments by advisor
- Standardize the prelim timing/content
- Provide timely decisions on TA/RA appointments
- Update student handbook

# Discussion Points…

***Survey Results:***

- **Need for updated graduate student handbook and structured PhD program guidelines**
- **Issues with timely TA/RA letter submissions and funding stability**
- **Disconnect between research labs**
- **Concerns about advisor-student relationships, including unreasonable demands and micromanagement**
- **Conflict Resolution:**
  - **Advisor → DGS → Dept. Head → Graduate College**